# A 13.56MHz RFID Transponder Front-End with Merged Load Modulation and Voltage Doubler-Clamping Rectifier Circuits

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***Abstract*—A 13.56 MHz RF transponder front-end circuit was designed for RFID or wireless sensor applications using 0.35-µm CMOS technologies. It converts RF power to DC and extracts clock and data; a smartly positioned NMOS device implements load modulation for efficient data sending back. The new design merges the load modulation and voltage clamping into the voltage doubler rectifier circuits to achieve longer operation distance. Simulation results show that the transponder operates over a wide range of electromagnetic field strength with induced voltage from 0.1V to 10V. The chip also generates a Power on Reset (POR) signal for digital circuits. The low power analog blocks consume less than 1µA current for the 0.1V induced voltage.**

## I. Introduction

In recent years, Radio Frequency Identification (RFID) technology has seen rapid growth in areas of supply chain management, access control, public transportation, airline baggage tracking, etc. A key low power transponder IC is needed for long-range operation [1]. In biomedical applications [2], an implanted passive transponder with sensors needs a similar RF interface to communicate with an external reader for two-way data transmission. In still other applications, e.g. a temperature logger, a temperature sensor with an A/D converter and EEPROM memory are used to convert and store temperature data, that are read periodically by a wireless reader. In such a situation, a battery is needed to operate the sensor and A/D converter. However, it is preferred for the RF interface to extract power and perform data communications by using the RF power for longer battery life.

Load modulation is the most frequently used technique for data transmission from transponder to reader [1-5]. A conventional load modulation approach is to use a load resistor in series with a MOS switch, and place them in shunt with the resonant circuit. However, the MOS switch cannot provide good RF isolation. As a result, the resistor and the MOS switch form an extra load to the resonant circuit. Thus, it requires stronger electromagnetic field strength to power up. This limits the operation distance between the transponder and the reader. To achieve longer distance operation, capacitive load modulation is generally used [3, 4]. In addition, voltage doublers have been used to boost the rectified DC voltage, so that the transponder can operate in weaker field. However, when the transponder is close to the reader, the field strength is strong and the voltage doubler will cause voltage overstress to the IC. This paper presents a new RF transponder front-end design that merges the load modulation and voltage clamping into the voltage doubler rectifier circuits to achieve longer operation distance. Simulation results show that the transponder operates over a wide range of electromagnetic field strength with induced voltage from 0.1V to 10V. The chip also generates a Power on Reset (POR) signal for digital circuits. The low power analog blocks consume less than 1µA current for the 0.1V induced voltage.

## II. Circuit Block Diagram

The chip operates based on the inductive coupling RFID principle [3]. Fig. 1 shows the block diagram of the RF front-end circuit. An off-chip inductor is used for inductive coupling; the rest of the front-end circuits are on-chip as indicated by the dotted rectangle. An RFID type reader transfers data and energy by sending a modulated RF wave using a coil antenna which couples with $L_1$, the external off chip coil inductor; $C_1$ is an on-chip capacitor that resonates with $L_1$ at 13.56MHz. An RF voltage is induced by the electromagnetic field strength and boosted by the resonant circuit to a level that can power up the chip after rectification. A voltage doubler rectifier circuit is used to convert the RF voltage to a DC voltage onto a capacitor. The data is low pass filtered and detected by a data comparator. The bias circuit generates a bias current for analog blocks while issuing a POR signal for any on-chip digital circuits. The clock is extracted using a simple Schmitt trigger circuit. In weak field condition, the chip starts to operate when the DC voltage reaches above 1.2V. However, in strong field case, both the RF and DC voltages are too high, causing potential damage to the circuit. To avoid this situation, RF and DC clamping circuits must be added to limit the chip operating voltage to a safe level. The system is designed to detect an Amplitude Shift Keying (ASK) signal with modulation index from 10% to 100% for a data rate of 100kbits/s at a carrier frequency 13.56MHz. The Data Modulation block is used to create a backward data modulation by changing the load of the RF resonant circuit for sending data back to the reader.

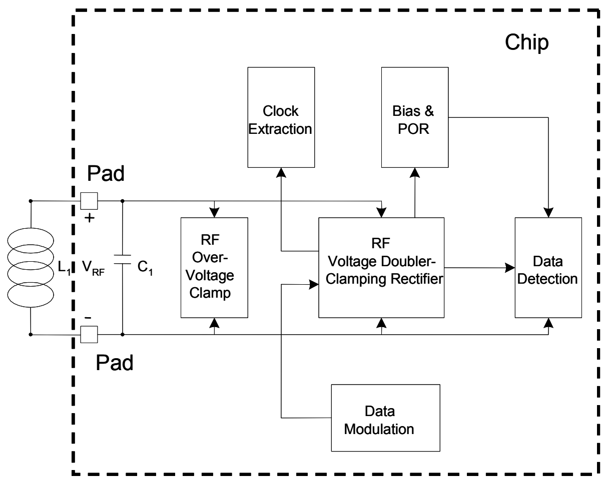



Fig. 1: RF front-end circuit block diagram

## III. CIRCUIT DESIGN

### A. RF Resonant and Clamp Circuits

Fig. 2 shows the main circuit signal path for the power conversion and data detection. $L_1$ and $C_1$ form an RF tank circuit with resonant frequency $f_0$, and quality factor Q as:

$$f_0 = \frac{1}{2\pi\sqrt{L_1 C_1}} \qquad (1)$$

$$Q = \frac{R_L}{2\pi f_0 L_1} = 2\pi f_0 R_L C_1 \qquad (2)$$

where $R_L$ is the equivalent load resistance of the transponder chip. The step-up RF voltage at the chip input is proportional to the circuit Q, which relates to the power consumption of the chip. The PMOS devices P1-P3, and P4-P6 provide a voltage clamping function when the electromagnetic field strength is strong. P1-P3 will clamp VRF in the positive direction while P4-P6 in the negative one. When these devices are on, they essentially conduct high currents and decrease the circuit Q, thus limiting the RF step-up voltage to a safe level.

### B. RF Rectification

The voltage doubler RF rectifier includes $C_2$, $C_3$, P11, and P7-P10. The converted DC voltage is on $C_3$, and is used as the power supply for the chip. P7-P10 is part of the rectifier and also acts as a further RF voltage clamp to limit the voltage at VRFD node to less than 5.0V for this 0.35um TSMC process. In layout, double guard rings are used for all the clamping PMOS devices.

### C. Bias and POR circuits

As shown in Fig. 3, the bias current is generated using the $V_t$ difference of two subthreshold biased NMOS N1 & N2. Devices P6, N7, and N8 form the start-up circuit. Transistor P6 is a long channel device optimized for low power consumption. The generated bias current, IBIAS, has a typical value around 100nA, which serves as the bias current for the data comparator. When the output DC voltage is above a certain level, a high to low transition POR signal is issued for all the digital latches. P3 and N5 need to be designed carefully to maintain a successful POR event. They are biased in strong inversion region. If they are on, all other devices in digital circuits will be operational. A good design criteria is to bias their Vgs to be 100mV above their threshold voltages.

### D. Data Detection

The data detection circuit demodulates the incoming ASK modulated signal, and this is completed by comparing the available DC voltage VDC, which still contains the modulated data component, with a low pass filtered version of it, $V_{TH}$. The low pass filter is simply implemented by a long channel PMOS P12 and a poly capacitor $C_4$. The data comparator CMP is conventional one and its schematic not shown here.

### E. Data Modulation

As mentioned earlier, capacitive load modulation using a capacitor in series with a MOS switch has been used to achieve longer transmission distance. Generally, they are placed in shunt with the resonant circuit before the voltage doubling capacitor $C_2$. This paper presents a new capacitive load modulation design. As shown in Fig. 2, the data modulation signal DATA_MOD controls the gate of MOS switch N1, which is placed within the voltage doubler rectifier circuit after the voltage doubling capacitor $C_2$. One of this switch device's body diode forms the reverse diode relative to VRFD and completes the voltage doubler circuits. Guard rings are also used for this device not to cause any potential latch up issue. When it is on, it shorts VRFD node to ground, abruptly dissipating RF energy from the resonant circuit. The result is that high modulation index is achieved with a quick time response. In addition, the voltage doubling capacitor, $C_2$, is reused as the capacitive load for load modulation. This approach reduces the number of capacitors.

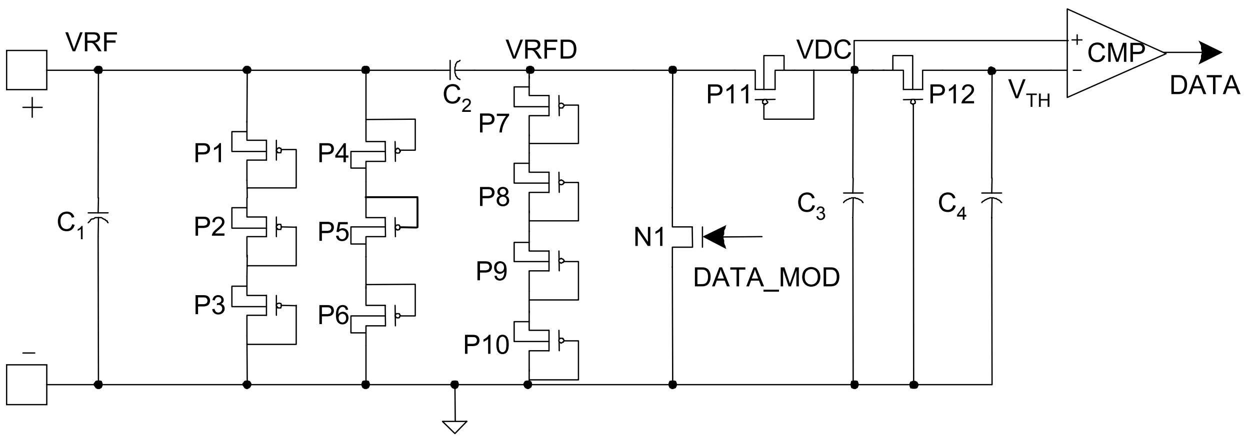



Fig. 2: RFID front-end circuit signal path schematic

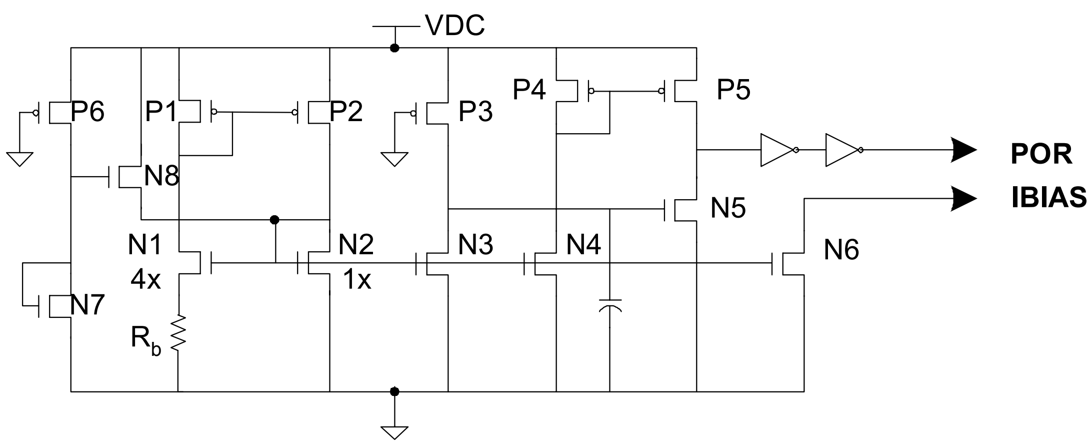



Fig 3: Schematic of the Bias and POR circuits

Table 1 lists the finished design component and device values; the capacitor $C_2$ in the design is about 40pF. A 40pF capacitor occupies quite significant die area; thus, this new design results much smaller die area.

TABLE I. DEVICE AND COMPONENT SIZES

| COMPONENTS | VALUE | DEVICES | W/L (μm) |
|---|---|---|---|
| $C_1$ | 34pF | P1-P10 | 50/0.5 |
| $C_2$ | 40pF | P11 | 100/0.5 |
| $C_3$ | 100pF | P12 | 0.8/40 |
| $C_4$ | 20pF | N1 | 2500/0.5 |

## IV. SIMULATIONS

### A. *Inductive Coupling Modeling*

The RFID interface has been designed and simulations run to verify its performance. An RF circuit model of the resonant circuit is required for the simulation purposes. For an inductive coupling RF interface, we can decouple the transponder coil from the reader antenna by introducing an induced voltage $U_i$ using equation 4.29 in reference [3] as shown in Fig. 4.

$$U_i = \mu_0 \cdot A \cdot N \cdot \omega \cdot H \quad (3)$$

$$V_{RF} = \frac{jU_i}{1 + (j\omega L_1 + R_1)(\frac{1}{R_L} + j\omega C_1)} \quad (4)$$

where $\mu_0$ is permeability constant, A is the cross section area of the transponder coil, N is the coil winding number, ω is the angular frequency, and H is the magnetic field strength. Using the data in [3], for A=0.048x0.076 $m^2$ (smart card antenna), $L_1$=3.6μH, N=4, $U_i$ is calculated to be 0.16V at a magnetic field strength H=0.1A/m, which is a typical value for an RFID reader at a distance of 80cm. Here, the design target is to operate the transponder to work at a minimum induced voltage of 0.1V. In this design, $L_1$ is set at 3.6μH to facilitate future hardware test.

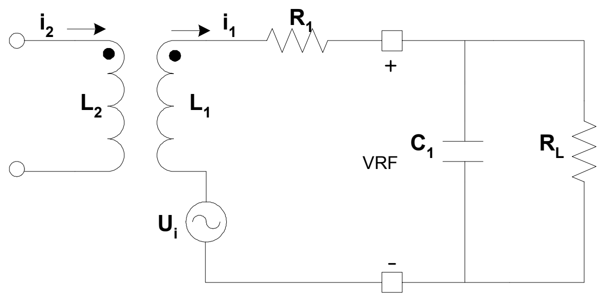



Fig. 4: Inductive coupling modeling

## B. Simulation Results

Fig. 5 shows the simulated data detection diagram. The input is 10% modulated ASK signal at an induced voltage level 0.1V peak and 0.08V valley. After the DC voltage is ready, POR transits from high to low. The data signal starts to give output after 200μs. This is due to slow response time of the bias current generation circuit. The VDC level is above 1.8V under this weak field condition. The bias and data detection circuits consume less than 1μA current.

The data modulation was simulated for weak and strong field conditions. Fig. 6 shows the results for an induced RF input voltage 0.1V and 10V, respectively. In weak field case as in (a), VRF undergoes almost 100% amplitude variation with a very fast response time. This means the circuit can work at even faster data rates. In (b), the incoming RF power is so strong that the clamping circuits are fully on. This makes it much harder to modulate data back. Although the amplitude modulation is not obvious from the diagram, it achieved 60mV peak amplitude change, which is above the minimum 10mV specification as in [6].

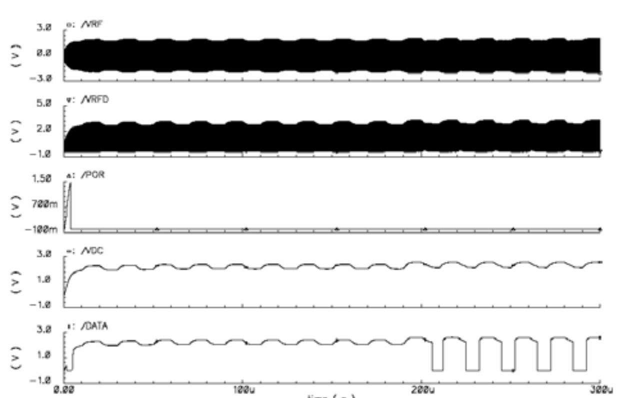
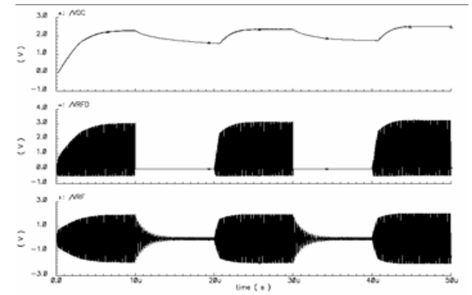
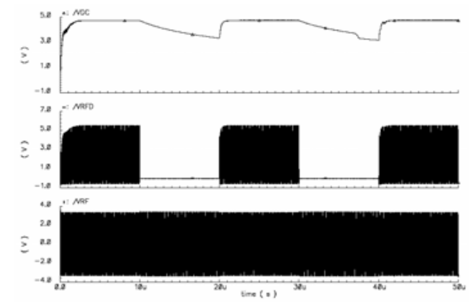


Fig. 5: Simulation plots of data detection at an induced voltage of 0.1V with 10% modulation index



(a) 0.1V induced RF voltage



(b) 10V induced RF voltage

Fig. 6: Data modulation simulations for two different induced RF input voltages

## V. CONCLUSIONS

A 13.56MHz RF front-end circuit has been designed and simulated. The new capacitive load modulation design is integrated with voltage doubler-clamping rectifier circuits, enabling the transponder to operate over wide range of electromagnetic field strength. Simulation results showed that the circuits function well for both ends of electromagnetic field strength with induced RF voltage from 0.1V to 10V. The low power analog blocks including the Bias & POR and Data Detection circuits consume less than 1μA current.

## REFERENCES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